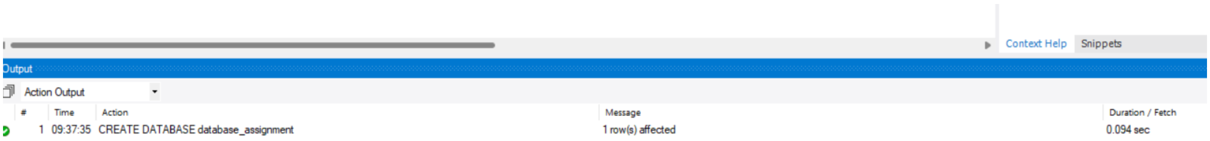
Step 1: Create Database

ANS :

CREATE DATABASE database_assignment;
USE database_assignment;



Step 2: Create Country Table

ANS :

CREATE TABLE country (

    id INT PRIMARY KEY,

    country_name VARCHAR(100),

    country_name_eng VARCHAR(100),

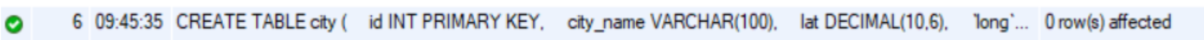
    country_code VARCHAR(10)

);

Step 3: Create City Table

ANS :

```
CREATE TABLE city (
   id INT PRIMARY KEY,
   city_name VARCHAR(100),
   lat DECIMAL(10,6),
   `long` DECIMAL(10,6),
   country_id INT,
   FOREIGN KEY (country_id) REFERENCES country(id)
);
```



Step 4: Create Customer Table

ANS :

```
CREATE TABLE customer (
   id INT PRIMARY KEY,
   customer_name VARCHAR(100),
   city_id INT,
   customer_address VARCHAR(100),
   next_call_date DATE,
   ts_inserted DATETIME,
   FOREIGN KEY (city_id) REFERENCES city(id)
);
```

7	09:45:39	CREATE TABLE customer (id INT PRIMARY KEY, customer_name VARCHAR(100), city_id INT, c...	0 row(s) affected

Step 5: Insert Data into Country

ANS : INSERT INTO country VALUES

(1,'Deutschland','Germany','DEU'),

(2,'Srbija','Serbia','SRB'),

(3,'Hrvatska','Croatia','HRV'),

(4,'United States of America','United States of America','USA'),

(5,'Polska','Poland','POL'),

(6,'España','Spain','ESP'),

(7,'Rossiya','Russia','RUS');

11	09:50:25	USE database_assignment	0 row(s) affected
12	09:50:25	INSERT INTO country VALUES (1,'Deutschland','Germany','DEU'), (2,'Srbija','Serbia','SRB'), (3,'Hrvatska','Croa...	7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0

Step 6: Insert Data into City

ANS : INSERT INTO city VALUES

(1,'Berlin',52.520008,13.404954,1),

(2,'Belgrade',44.787197,20.457273,2),

(3,'Zagreb',45.815399,15.966568,3),

(4,'New York',40.730610,-73.935242,4),

(5,'Los Angeles',34.052235,-118.243683,4),

(6,'Warsaw',52.237049,21.017532,5);

	#	Time	Action	Message
✔	15	09:51:13	USE database_assignment	0 row(s) affected
✔	16	09:51:13	INSERT INTO city VALUES (1,'Berlin',52.520008,13.404954,1), (2,'Belgrade',44.787197,20.457273,2), (3,'Zag...	6 row(s) affected Records: 6 Duplicates: 0 Warnings: 0

Step 7: Insert Data into Customer

ANS : INSERT INTO customer VALUES

(1,'Jewelry Store',4,'Long Street 120','2020-01-21','2020-01-09 14:01:20'),

(2,'Bakery',1,'Kurfürstendamm 25','2020-02-21','2020-01-09 17:52:15'),

(3,'Café',1,'Tauentzienstraße 44','2020-01-21','2020-01-10 08:02:49'),

(4,'Restaurant',3,'Ulica Ipa 15','2020-01-21','2020-01-10 09:20:21');

	#	Time	Action	Message
✔	17	09:51:25	USE database_assignment	0 row(s) affected
✔	18	09:51:25	INSERT INTO customer VALUES (1,'Jewelry Store',4,'Long Street 120','2020-01-21','2020-01-09 14:01:20'), (2,...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

Step 10: Display Customer Table

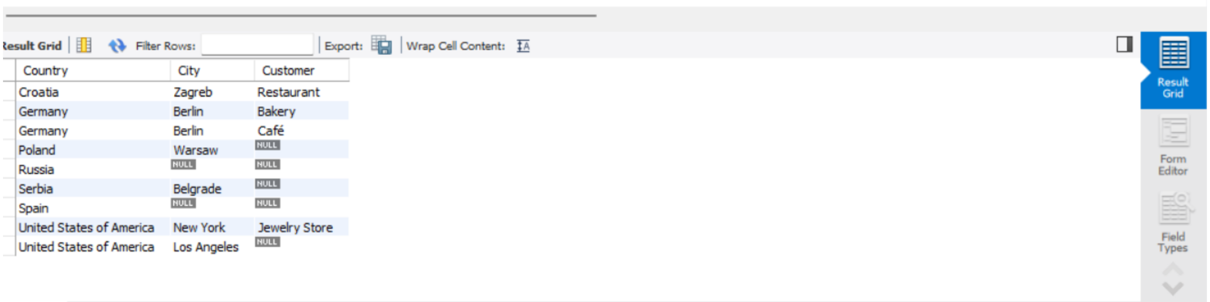
ANS : Step 10: Display Customer Table

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	customer_name	city_id	customer_address	next_call_date	ts_inserted
1	Jewelry Store	4	Long Street 120	2020-01-21	2020-01-09 14:01:20
2	Bakery	1	Kurfürstendamm 25	2020-02-21	2020-01-09 17:52:15
3	Café	1	Tauentzienstraße 44	2020-01-21	2020-01-10 08:02:49
4	Restaurant	3	Ulica Ipa 15	2020-01-21	2020-01-10 09:20:21
NULL	NULL	NULL	NULL	NULL	NULL

Task 1 : (join multiple tables using left join) List all Countries and customers related to these countries. For each country displaying its name in English, the name of the city customer is located in as well as the name of the customer. Return even countries without related cities and customers.

ANS :
```
SELECT
   c.country_name_eng AS Country,
   ct.city_name AS City,
   cu.customer_name AS Customer
FROM country c
LEFT JOIN city ct
ON c.id = ct.country_id
LEFT JOIN customer cu
ON ct.id = cu.city_id
ORDER BY c.country_name_eng;
```



Country	City	Customer
Croatia	Zagreb	Restaurant
Germany	Berlin	Bakery
Germany	Berlin	Café
Poland	Warsaw	NULL
Russia	NULL	NULL
Serbia	Belgrade	NULL
Spain	NULL	NULL
United States of America	New York	Jewelry Store
United States of America	Los Angeles	NULL

Task 2 : (join multiple tables using both left and inner join) Return the list of all countries that have pairs(exclude countries which are not referenced by any city). For such pairs return all customers. Return even pairs of not having a single customer

ANS : SELECT

```
    c.country_name_eng AS Country,
    ct.city_name AS City,
    cu.customer_name AS Customer
FROM country c
INNER JOIN city ct
ON c.id = ct.country_id
LEFT JOIN customer cu
ON ct.id = cu.city_id
ORDER BY c.country_name_eng;
```

Country	City	Customer
Croatia	Zagreb	Restaurant
Germany	Berlin	Bakery
Germany	Berlin	Café
Poland	Warsaw	NULL
Serbia	Belgrade	NULL
United States of America	New York	Jewelry Store
United States of America	Los Angeles	NULL